AUTOSAR HIGH-LEVEL DESIGN SPECIFICATION
Document: AUTOSAR_HLD_Sample_BrakingSystem
Classification: Automotive Technical Specification

1.0 Architecture Overview
The SWC_BrakingController is responsible for vehicle deceleration arbitration.
It interacts with chassis sensors and powertrain actuators.
The subsystem manages anti-lock braking modulation, hydraulic pressure blending, and regenerative braking command coordination.

2.0 Port Specifications for SWC_BrakingController

The following interface ports are defined for the component:

Port Name	Direction	Interface	Description
P_BrakePressure	Provided	If_BrakePressure	Current hydraulic pressure
R_PedalPosition	Required	If_PedalPosition	Driver brake demand

Notes: P_BrakePressure broadcasts sensor readings at 10ms periodic rate.
R_PedalPosition receives redundant sensor readings for ISO 26262 ASIL-D safety integrity.